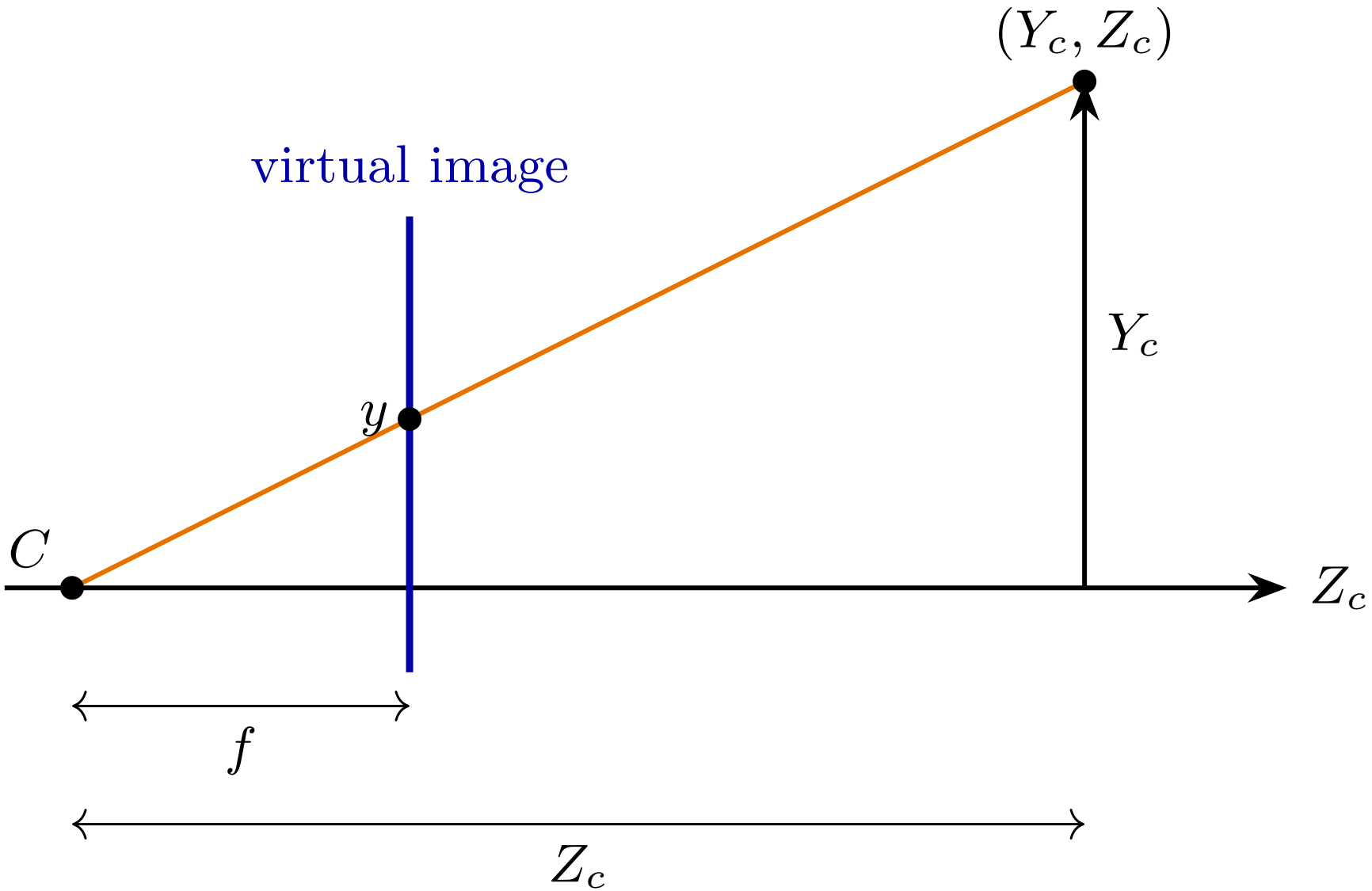Side view: $y/f = Y_c/Z_c$.
The physical sensor behind the pinhole has inverted signs.